Solution Requirements

Date	7 July 2026
Team ID	xxxxxx
Project Name	EduGenie Google Gemini Powered Learning Assistant
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Users (Students, Self-learners, Educators) must securely log in, sign up, or verify their identity using standard credentials to access personalized profiles.	High
2	Authorization levels	The system must distinguish between user roles (e.g., Student vs. Educator) to grant appropriate permissions, such as viewing vs. managing quiz templates.	Medium
3	External interfaces	The system must connect seamlessly to the Google Generative AI SDK (Gemini 1.5	High

		Pro) for advanced cognitive tasks and interface locally with the LaMini-Flan-T5 model weights via the Transformers library	
4	Transactions processing	The application must process incoming learning requests sequentially, structured as standard HTTP POST payloads, managing asynchronous interactions with cloud AI endpoints safely without losing state.	High
5	Reporting	The platform must generate real-time feedback summaries, interactive quiz performance breakdowns (correct vs. incorrect choices), and structured markdown-based multi-level roadmaps for user self-evaluation.	Medium
6	Business rules, etc.	AI prompt handling must enforce input string length boundaries (matching hardware safety constraints) and prevent model hallucination by anchoring prompt engineering context to strict educational rules.	High
7	Compliance to laws or regulations	The platform must apply content safety filters within the AI prompt engineering pipeline to ensure generated educational material complies with standard age-appropriate safety guidelines.	Medium

8	*Other*	The platform must support offline template rendering via Jinja2 so that structural web assets load immediately even if the external network is unstable.	Low

Step 2: Non-Functional Requirements (NFR)

S.No	**NFR Category**	**Requirement Description**	**Target Metric / Acceptance Criteria**
1	Performance &	The local text processing	e.g., Basic text features process in < 2 seconds

	Speed	pipeline using LaMini-Flan-T5 must complete structural processing and basic content text validation within 2 seconds.	
2	Scalability	The asynchronous FastAPI framework must handle concurrent learning sessions smoothly and process incoming API requests without dropping tasks.	e.g., Handles simultaneous user query streams efficiently
3	Security & Data Privacy	Third-party cloud integration tokens (Google Gemini API Key) must be strictly isolated via secure server environment variables to prevent credentials exposure.	e.g., Local environment variables isolation
4	Reliability & Availability	The local FastAPI application container and system endpoints must remain operational without crashes during runtime, gracefully managing external API connection drops.	e.g., 99.5% local application uptime during usage sessions
5	Usability & Accessibility	The HTML + CSS frontend interface must feature high-contrast text elements, intuitive task navigation controls, and fully responsive layouts across devices.	e.g., Complies with standard responsive and accessible design patterns
6	*Other*	The system source code must maintain a clean modular architecture to ensure cross-platform compatibility across Windows, Linux, and macOS platforms	e.g., Cross-platform software portability